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Review article

Factors predicting post-stroke aphasia recovery

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ABSTRACT

Background: Aphasia is an important stroke sequel that impacts negatively on the HQoL of stroke patients. Although a number of stroke patients with aphasia will have good functional recovery, many are left with language deficits.
Method: Papers were identified through PubMed and MEDLINE search, with keywords such as: 'stroke', 'aphasia', 'post-stroke aphasia', 'factors that predict aphasia recovery', 'aphasia outcomes' and 'aphasia prognosis'.
Results: The most important factors that determine recovery are the lesion location and size, aphasia type and severity and to some extent the nature of early haemodynamic response, and treatment received. Anagraphic factors like gender, age, handedness and education have not been found to be robust predictors of recovery.
Conclusions: Predicting post-stroke aphasia recovery is difficult, because of the interplay between lesion, anagraphic, and treatment-related factors, in addition to the role of neuroplasticity.

Contents

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1. Introduction

Aphasia is an acquired disorder of language processing which may affect speech comprehension, expression, writing or reading; leaving other cognitive capacity intact [18]. Aphasia in adults is most often secondary to stroke in the language-dominant cerebral hemisphere [13], present in 20% to 40% of acute stroke patients [32,78]. The incidence of post-stroke aphasia ranges from 40 to 60 per 100,000 per annum [22,24]. Roughly 250,000 persons live with aphasia in the U.K. [81]. Post-stroke aphasia is associated with more severe strokes, higher mortality, decreased rate of functional recovery, and higher health-care cost [22,23]. Patients are more likely to seek medical help earlier, and to receive thrombolysis compared to patients with no aphasia [24,74].

Method/Literature search

Papers for this review were identified through searches of PubMed and MEDLINE, using the following keywords: 'stroke', 'aphasia', 'post-stroke aphasia', 'factors that predict aphasia recovery', 'aphasia outcomes' and 'aphasia prognosis'. The reference lists of other review papers were used to provide further leads.

After a stroke, aphasic patients show some degree of gradual spontaneous recovery in the first few months [25,62]; however many post-stroke aphasic patients are left with some chronic deficit [56]. Since language is important for interpersonal relationships, these patients have significantly lower health-related quality of life (HQoL) with reduced probability of returning to work and social activities [10].

Aphasia recovery can be influenced by lesion-related factors such as the site and extent of the lesion; the type and severity of initial aphasia; and inter-individual variability in brain network. Non-lesional or anagraphic factors such as age, gender, handedness, and environmental characteristics may influence recovery [6]. In addition speech therapy is now considered to also predict aphasia recovery [9,82]. The above factors depend significantly on the degree of early reperfusion, recovery from diaschisis, neuroplasticity and language reorganisation [35,45].

This review aims to discuss factors that influence post-stroke aphasia recovery.

2. Factors predicting better or worse recovery

Post-stroke aphasia recovery involves rebuilding neural circuitry for language, which may depend on a number of lesion-related, non-lesion related and treatment-related factors discussed below.

2.1. Lesion (stroke) related factors

2.1.1. Lesion location and size

Studies have reported the negative influence of larger lesion on post-stroke aphasia recovery; and considered lesion size to be an important predictor of recovery [29,62,67,70,71,73]. Aphasia recovery is generally predicted not only by size but also by lesion location in critical language areas, logically a small lesion in a speech area is more likely to impact on language severity and recovery, while a large lesion elsewhere may affect speech minimally [38,70,73].

Lesions in the superior temporal gyrus (STG) produce a more persistent global aphasia, which is associated with poor aphasia recovery [19,34,50,76,90], especially the posterior part of the STG [1,50,72]. Cortical lesions tend to have severe aphasia compared to subcortical lesions; as such subcortically located aphasias often have a better prognosis [48].

Apart from a study by Lazar et al. [58] reporting that lesion size does not predict language recovery; more studies attest that recovery is inversely related to the size of lesion, with a preserved left superior temporal gyrus as one of the most important factor for satisfactory recovery, and an intact basal ganglia also contributing to better recovery [35,38].

2.1.2. Stroke and aphasia severity

A number of studies observed that initial aphasia severity impacts negatively on aphasia recovery [4,26,56,57,77,78]. Maas et al. [62] noted that a lower initial NIHSS score was associated with a favourable outcome and about 90% of aphasia patients with NIHSS scores < 5 recover fully without receiving thrombolysis. Pedersen et al. [77] reported that the only clinically relevant predictor of aphasia was the initial severity of aphasia. Better improvement was also observed in patients with less affected initial speech function in a study by Laska et al. [56]. A contrary view was observed by Seniów et al. [92] and Lazar et al. [58] that stroke severity does not predict language outcome. Despite the inferences made that the initial aphasia severity may serve as a useful predictor of outcome, it will be difficult to draw a strong conclusion from these studies, since these studies did not tightly control for initial severity, lesion site and lesion size. In addition these studies did not also control for the roles of various non-linguistic cognitive abilities, like motivational, emotional, and social characteristics.

2.1.3. Type of language deficit

Studies have reported lower rates of recovery in global and anomic aphasia and better recovery in Broca's and conduction aphasia; with global aphasia patients more likely to still have deficits a year later [20,51,78].

Jung et al. [46] observed that those with global aphasia had poorer recovery than those with other aphasia types, which may reflect higher stroke severity. Global aphasics are typically more impaired in all tasks. The authors noted that initial severity of aphasia was strongly associated with improvement in speech function, with lower initial severity showing good responses. A study by Mazzoni et al. [67] reported that speech comprehension had a better recovery compared to expression, while Basso et al. [4] reported that transcortical sensory aphasia had a worse prognosis than Broca's or transcortical motor aphasia. On the contrary Sarno & Levita [88] found no relationship between aphasia recovery and aphasia type. El Hachioui et al. [32] noted that different linguistic modalities appear to recover at different times, with phonology improving earlier than semantic or syntactic language, and language expression recovering later than receptive language.

2.1.4. Stroke subtype

Studies by Basso et al. [3,5] and Jung et al. [46] reported that haemorrhagic stroke survivors had a better prognosis than ischaemic stroke patients, suggesting that ischaemic stroke is a negative predictor for aphasia recovery. The better prognosis may be due to fibre bundles being displaced without damage in haemorrhagic strokes [6].

2.1.5. Metabolic factors

Better aphasia recovery is determined by the normal return of cerebral blood flow (CBF), the degree of successful reperfusion, and the return of normal glucose metabolism of infarcted tissue and perilesional areas [27,36,49]. Richardson et al. [83] reported a less clear relationship between CBF measured at baseline and the success

Table 1
Lesion related factors.

Lesion size	Larger lesions predict poor aphasia recovery.	Naeser et al. [70,73]; Mazzoni et al. [67]; Goldenberg & Spatt [29]; Naeser & Palumbo [71]; Heiss et al. [35]; Maas et al. [62]; Henseler et al. [38]
	Lesion size does not predict language recovery.	Lazar et al. [58]
Lesion location	Lesions in the STG predict poor aphasia recovery.	Kertesz et al. [50]; Hanlon et al. [34]; Demeurisse & Capon [19]; Parkinson et al. [76]
	Lesions located in the posterior two thirds of the STG predict poor aphasia recovery.	Naeser et al. [72]; Kertesz et al. [50]; Alexander et al. [1]
	Anterior–inferior lesions in the STG are associated with poorer comprehension recovery.	Selnes et al. [90]
	Subcortical located lesions have a better prognosis than cortical lesions.	Kang et al. [48]
Aphasia/stroke severity	Initial stroke/aphasia severity impacts negatively on aphasia recovery.	Basso [4]; Pedersen et al. [77]; Laska et al. [56]; Pedersen et al. [78]; Fillingham et al. [26]; Lazar et al. [57]; Maas et al. [62]
	Stroke severity does not predict language outcome.	Seniów et al. [92]; Lazar et al. [58]
Type of speech deficit	Global aphasia and anomic aphasia have lower rates of recovery.	Kertesz & Mccabe [51]; Demeurisse et al. [20]; Pedersen et al. [78]; Jung et al. [46].
	Speech comprehension has better recovery compared to expression.	Mazzoni et al. [67]
	Transcortical sensory aphasics have a worse prognosis than Broca's or transcortical motor aphasics.	Basso et al. [4]
	Phonology improves earlier than semantic or syntactic language.	El Hachioui et al. [32]
	No relationship between aphasia recovery and aphasia type.	Sarno & Levita [88]
Stroke subtype/pathology	Haemorrhagic stroke has better prognosis than ischaemic stroke.	Basso et al. [3,5] and Jung et al. [46]
Metabolic factors	Early return of normal CBF and CMR predict better language recovery.	Karbe et al. [49]; Fridriksson et al. [27]; Richardson et al. [83]; Mimura et al. [68]; Hillis [40]

Summary: There is evidence to suggest that larger lesions, STG located lesions, initial aphasia severity, and metabolic factors predict recovery in patients with post-stroke aphasia.

of anomia treatment. A study by Mimura et al. [68] noted that an early recovery of left hemisphere CBF correlated with lessened initial impairment of language and better recovery within a year. In summary, a continuing hemisphere hypoperfusion and the delay in return of normal CBF and cerebral metabolic rate (CMR) adjacent to lesions during the early post-stroke period lead to poorer long-term recovery [40] [see Table 1 for summary].

2.2. Non-lesion (patient) related factors

2.2.1. Gender

The incidence of aphasia is reported to be higher among females in some studies [3,5,24,39,55]; higher among males in a study by Kertesz & Sheppard [52], while Kang et al. [48] reported no gender variation.

Basso et al. [3,5] reported that females recover significantly better in oral expression than males; while a study by Pizzamiglio et al. [80] reported that females with global aphasia had significantly better improvement in language comprehension. Studies by Pedersen et al. [77], Lendrem & Lincoln [61], Inatomi et al. [44], Seniów et al. [92], and Godefroy et al. [28] reported no gender difference in aphasia recovery, and that gender differences in aphasia type did not appear to influence aphasia recovery. In summary there is weak and inconclusive evidence that gender predicts functional recovery from aphasia.

2.2.2. Age

Generally patients with aphasia are more likely to be older and report a higher prevalence of fluent aphasia [24,77,78]; while De Renzi et al. [21] reported Broca's aphasia to be commoner in younger patients. Kang et al. [48] found no difference between age and aphasia type or severity.

A study by Pickersgill & Lincoln [79] found that younger patients with aphasia improved better than older patients. Lendrem & Lincoln [61] found an influence of younger age on recovery in those treated for aphasia. Laska et al. [56] reported that older age is a negative predictor for improvement. The studies by Kertesz & Mccabe [51] and Pedersen et al. [77] reported a non-significant age difference. Other studies found no influence of age on aphasia recovery [41,44,58,61,78,92]. The influence of age on aphasia recovery still remains unclear, with a tendency for older patients to have a poorer recovery.

Table 2
Non-lesion related factors.

Gender	Females recover better than males in oral expression	Basso et al. [3,5]
	Females with global aphasia had better improvement	Pizzamiglio et al. [80]
	No gender difference in aphasia recovery	Pedersen et al. [77]; Lendrem & Lincoln [61]; Inatomi et al. [44]; Seniów et al. [92]; Godefroy et al. [28]
Age	Older age predicts poorer aphasia recovery	Pickersgill & Lincoln [79] Lendrem & Lincoln [61] Laska et al. [56]
	Non-significant tendency towards an age influence	Kertesz & Mccabe [51]; Pedersen et al. [77]
	Age does not predict aphasia recovery	Lazar et al. [58]; Lendrem & Lincoln [61]; Holland et al. [41]; Inatomi et al. [44]; Seniów et al. [92]; Pedersen et al. [78]
Handedness	No influence of handedness on aphasia recovery	Pedersen et al. [77]; Pickersgill & Lincoln [79]
Preexisting cognitive deficits	A correlation between cognitive decline and aphasia recovery, which was stronger in post-traumatic aphasia	Vukovic et al. [96]
	The role of cognitive deficit remains unclear	Leff et al. [60]
Education	A preserved visuo-spatial working memory is associated with better improvement in comprehension and naming	Seniów et al. [92]
	No relationship between educational attainment and aphasia recovery	Connor et al. [15]; Lazar et al. [58]

Summary: There is weak evidence to suggest that age and gender predict recovery in patients with post-stroke aphasia.

Table 3
Treatment related factors.

Treatment	Aphasia treatment positively influences recovery.	Mazzoni et al. [67]; Marsh et al. [64]; Rose et al. [86]; Fridriksson et al. [27]
	rTMS and tDCS are beneficial in improving aphasia recovery.	Holland et al. [42]; Marcotte et al. [63]; Volpato et al. [95]; Allendorfer et al. [2]; Kakuda et al. [47]; Naeser et al. [69]
	MIT, CID, AOT, and SFA may influence recovery.	Marcotte et al. [63]; Martins et al. [66]; Conklyn et al. [14]; Koenig-bruhin et al. [54]; Pulvermüller et al. [82]
	Donepezil and galantamine have shown some beneficial effects compared with placebo.	Berthier et al. [8]; Hong et al. [43]
	No influence of treatment on aphasia recovery.	David et al. [17]
	Combining speech therapy and tDCS had better outcome.	Jung et al. [46]
	Non-significant tendency for a better outcome among those on intensive compared to regular therapy.	Martins et al. [66]; Cherney et al. [12]

2.2.3. Handedness

Studies have not shown any influence between handedness and aphasia recovery despite the potential greater neural capacity for recovery in left-handers and the ambidextrous, since they are more likely to have language representation in both hemispheres and a higher likelihood for right hemisphere dominance [25,77,79].

2.2.4. Pre-existing cognitive deficits

A study by Vukovic et al. [96] observed a correlation between cognitive decline and poor aphasia recovery, which was weaker in post-stroke patients compared to post-traumatic patients with language deficit. Cognitive deficits in patients with post-stroke aphasia may not be related directly with functional recovery, but may influence learning during rehabilitation [26,85]. Therefore the role of cognition in post-stroke aphasia recovery remains unclear [60].

2.2.5. Educational attainment

Those with more years of education are less vulnerable to language disruption by stroke [15,30,59]. Seniów et al. [92] reported that an intact visuo-spatial memory correlated with better improvement in comprehension and naming. Other studies reported no influence of educational attainment on aphasia recovery [15,25,58].

2.2.6. Environmental factors — family support and motivation

Although the influence of environmental factors on aphasia recovery has not been widely studied [25]; it is generally considered that a highly supportive environment improves the outcome of patients with aphasia especially with regard to the effectiveness of therapy [11,54]. Stroke patients who are generally aware of their handicap and receive good support show more motivation and are more likely to have a better outcome [6] [see Table 2 for summary].

2.3. Treatment related factors

The influence of language therapy is difficult to assess because spontaneous recovery occurs in many patients [16]; patients with mild aphasia are usually excluded from studies to avoid the 'ceiling effect' [68]; and those under treatment are more likely to have severe speech deficits [9].

There is growing evidence that treatment of aphasia may positively influence recovery [27,64,67,86]. Studies on repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDCS) have affirmed their beneficial effects on post-stroke aphasia recovery by inducing neuroplasticity of targeted brain areas [2,42,47,63,69,95]. Therapeutic techniques like melodic intonation therapy (MIT), constraint induced therapy (CID), action observation treatment (AOT), and semantic feature analysis (SFA) tend to also facilitate neuroplasticity [14,54,63,66,82]. Pharmacological trials of donepezil and galantamine have shown some beneficial effects compared with placebos, as these drugs promote reorganisation and increase in acetylcholine concentration [8,43]. Piracetam, dextroamphetamine, dopamine and antidepressant trials have yielded mixed results [35,65].

Jung et al. [46] reported that a combination of speech therapy and tDCS had better outcome in those who commenced treatment earlier. A meta-analysis by Robey [84] reported that intensive therapy was more effective when started early. The SP-I-R-IT study [66] suggests better recovery for those on intensive compared to regular therapy. Weak evidence on the influence of treatment on aphasia recovery was reported in a study by David et al. [17] and a systematic review by Cherney et al. [12].

In summary, the type, duration, intensity and earlier commencement of aphasia therapy may be considered a determinant factor for better recovery [17]. The 'Rotterdam Aphasia Therapy Study-3' when concluded may give a better insight into the role of early-intensive cognitive therapy compared to regular language therapy [75] [for summary see Table 3].

3. The use of neuroimaging in predicting speech recovery/neuroplasticity

Advances in neuroimaging have led to new insights into the understanding of neuroplasticity. Evidence shows that aphasia recovery relies on activation of the perilesional and other redundant areas with potential for language function [33,53,89,91]. This 'redundancy recovery' is determined by the extent to which speech deficits are compensated

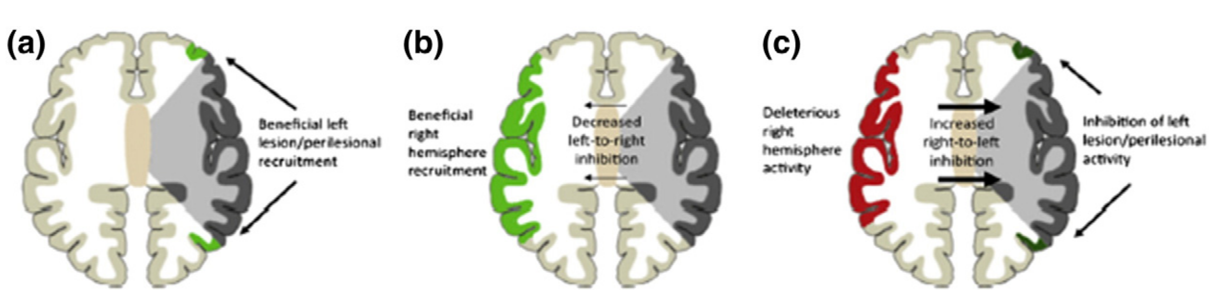

Fig. 1. The hierarchical model for neuroplasticity. (a) After a left hemisphere stroke (grey) there is recruitment of perilesional area (green). (b) Recruitment of right perisylvian area (grey) to subserve some language function. (c) Right hemisphere areas (red) may be deleterious impeding functional recovery of left hemisphere area.
Adapted from Hamilton et al. [33].

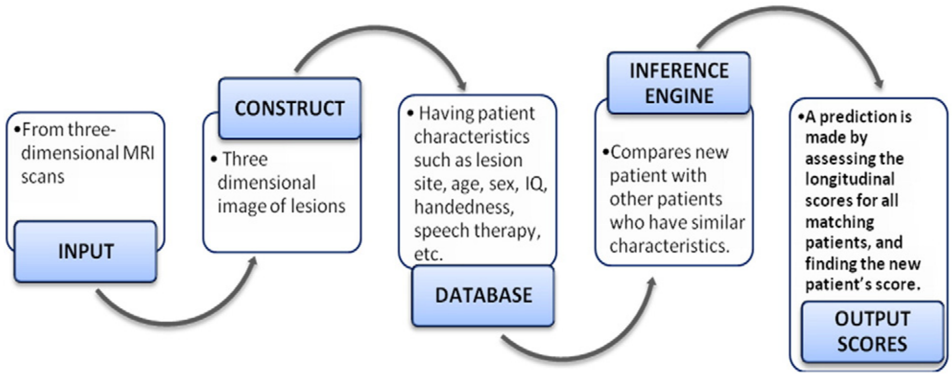

Fig. 2. Procedure and database for estimating language recovery. The input is a high resolution MRI scan of a new patient. The lesion is converted to a 3-dimensional image. The lesion is compared with that of other patients already in the database. The language scores of previous patients are plotted over time, enabling time course for the new patient to be estimated. The output will also depend on a variety of factors discussed above.
Modified from the PLORAS study [81].

for by networks of areas through release from inhibition of diaschisis [37,97].

There is now a hierarchical model for aphasia recovery; in which a patient with small left hemisphere lesion usually recovers well due to normal restitution of perilesional language networks; those with larger lesions, show recruitment of other areas of the left hemisphere surrounding the lesion (this restoration of function albeit incomplete may be satisfactory); if the ipsilateral hemisphere is severely damaged there will be activation of the homologous right hemisphere [33,37,97] [**see** Fig. 1].

The nature of the contribution from the contralateral hemisphere is being debated [35,45]. Some studies suggest that activation of the right hemisphere is less efficient [49,87], and others believe that its function is indistinct [94]. In summary, a good recovery can be anticipated when perilesional left hemisphere areas compensate for damaged language regions, while limited recovery is achieved when the right hemisphere is recruited for language tasks [93,97].

3.1. Predicting aphasia recovery

Predicting post-stroke aphasia recovery is important, as patients and their caregivers may want to know the chances of recovery [81]. Lazar et al. [57] developed a prediction model using the Western aphasia battery (WAB), that by 90 days, patients would have improved to approximately 70% of their maximum potential recovery. Predicting aphasia recovery is difficult due to substantial variability in outcomes even between two individuals with similar lesions [27,53]. The PLORAS study [81] in trying to address some of the complexities of predicting aphasia recovery, proposed a data-led predictive model for language outcome based on the recovery of previous patients with similar lesion and clinical characteristics. The model requires a large data pool of stroke patients. Using high resolution MRI scans the lesion is imaged in three-dimensions and compared with those of other patients in the database. The language scores and improvement of those who are most similar to the new patient are then extracted [Fig. 2].

Some of the problems that may be encountered with using these models of prediction include: (i) Patterns of deficit in patients are more likely to be complex, therefore restraining new patients within a locus of other patient's impairments may be fraught with errors [81]. (ii) Statistical complexities may arise from interplay between many factors. As the number of observations and predictors increases it makes statistical inferences difficult. This 'curse of dimensionality' results from difficulties in analysing increasing numbers of predictors [7]. Greenhouse et al. [31] observed that assessing relationships between variables and the probability of language recovery, a regression analysis needs to adjust for the effects of confounding variables before developing a model based on comparing observed probability of language recovery. (iii) The statistical complexities may also lead to posterior (P)-hacking; a tendency to manipulate the process of statistical analysis to get a statistically significant figure [81].

In conclusion, despite the number of stroke patients with aphasia having a good functional recovery, a large number are left with varying degrees of language deficits. Predicting post-stroke aphasia recovery is difficult, because of the interplay between lesion-, anagraphic-, and treatment-related factors, in addition, most of these studies were not tightly controlled for potential confounders. However, the most robust predictors of recovery appear to be lesion related factors.

Conflict of interest

We declare that we have no conflict of interest.

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